

citemsp Package

Per-Key Citation Locators

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Abstract

The `citemsp` package provides a single user-facing command, `\citemsp`, that attaches section (§), paragraph (¶), and prefix-based locators directly to numeric citation labels. It works on top of either `biblatex` or `natbib`, auto-detecting which backend is loaded. An extensible prefix registry currently provides ten built-in types, and users can register custom prefixes with `\citemspprefix`. The package parses a slash-delimited syntax, rendering each locator as a superscript/subscript pair on the corresponding citation number, producing output such as $[1_{5}^{\S 3.2}, 2_{pp. 520}^{pp. 520}, 3_{eq. 4.3}^{\S 1}]$. Plain entries (no slashes) are batched and passed to the backend's native `\cite`, so range compression (`numeric-comp`, `sort&compress`) works naturally.

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1 Motivation

The `biblatex` package offers a locator interface through its postnote field. A user can write `\autocite[§3.2]{smith2003}` to append a free-form locator after the citation number. For multi-key citations, `\autocites` accepts per-key pre- and post-notes:

<code>\autocite[§3.2]{smith2003}</code>	$\longrightarrow$	<code>[1, §3.2]</code>
<code>\autocites[§3.2]{smith2003}[¶2]{meyerrochow2003}</code>	$\longrightarrow$	<code>[1, §3.2, 3, ¶2]</code>

This approach has two primary limitations:

1. **Verbose syntax.** Each additional locator requires a full optional-argument pair, both in the source (`\autocites[...]{key}[...]{key}...`) and in the compiled output, where the postnote is typeset as plain bracketed text indistinguishable from the citation number. As the example above shows, the bracket grows linearly with every additional key and locator, quickly cluttering the running text.
2. **No registry, in either backend.** `natbib` and `biblatex` are the two general-purpose citation backends used by nearly all modern LaTeX documents. `natbib`'s postnote is free text with no type system at all. `biblatex` comes closer with a `pagination` field, but it recognizes only a small, localized set of generic types (page, chapter, section, paragraph, verse, line, column), is fixed once per bibliography entry rather than chosen per citation, and excludes scholarly types such as equation, theorem, figure, or lemma.

`citemsp` addresses both limitations directly while working on top of `biblatex` or `natbib`. Locators are rendered as a superscript/subscript pair, visually set apart from the citation number, and every key with its locators is passed in a single comma-separated argument rather than nested optional arguments. Ten built-in prefix types, extensible via `\citemspprefix`, give authors a consistent, per-citation locator registry that neither backend provides on its own.

2 Installation

2.1 Per-project

Copy `citemsp.sty` into the same directory as your main `.tex` file.

2.2 System-wide (local texmf)

```
TEXMFHOME=$(kpsewhich -var-value=TEXMFHOME)
mkdir -p "$TEXMFHOME/tex/latex/citemsp"
cp citemsp.sty "$TEXMFHOME/tex/latex/citemsp/"
texhash "$TEXMFHOME"
```

`TEXMFHOME` resolves to `~/texmf` on a vanilla TeX Live install (e.g., Linux) and to `~/Library/texmf` on MacTeX — using the `kpsewhich`-derived value keeps a single command portable across both.

3 Usage

3.1 Loading and compiling with biblatex

The package must be loaded *after* `biblatex`:

```
\usepackage[style=numeric, backend=biber]{biblatex}
\addbibresource{refs.bib}
\usepackage{citemsp}
```

Build with `latexmk -pdf <file>.tex`, or manually with the chain `pdflatex → biber → pdflatex → pdflatex`.

Running plain `bibtex` against a `backend=biber` document leaves the bibliography empty, since `biblatex/biber` write a different auxiliary-file format than classic `bibtex` expects.

3.2 Loading and compiling with natbib

```
\usepackage[numbers]{natbib} % omit if your class loads it
\usepackage{citemsp}
```

Build with `latexmk -pdf <file>.tex`, or manually with the chain `pdflatex → bibtex → pdflatex → pdflatex`.

Classes that auto-load `natbib` (e.g., `revtex4-2`, `aastex631`) require no explicit `\usepackage{natbib}`. Loading `citemsp` before either backend produces a package error.

3.3 Syntax

The `\citemsp` command accepts a comma-separated list of entries. Each entry has the form `key`, `key/loc1`, or `key/loc1/loc2`. By default, the first slot renders as § (section) and the second as ¶ (paragraph). A single-letter prefix changes the locator type:

<code>\citemsp{key}</code>	→	$[1]$
<code>\citemsp{key/section}</code>	→	$[1^{\S 6.2}]$
<code>\citemsp{key/section/paragraph}</code>	→	$[1^{\S 6.2}_{\P 3}]$
<code>\citemsp{key//paragraph}</code>	→	$[1_{\P 3}]$
<code>\citemsp{key/section/e4.3}</code>	→	$[1^{\S 6.2}_{\text{eq. 4.3}}]$

Note the subscript-only form `key//par`: an empty first slot with a double slash gives a paragraph (or prefixed) subscript without a section superscript.

3.4 Prefix registry

All ten built-in prefixes are listed below. Note that no prefix defaults to § (section) in the superscript slot and ¶ (paragraph) in the subscript slot. A prefix is a single letter at the start of a locator slot that selects a symbol:

Users can register custom prefixes with `\citemspprefix`. Place the command in the preamble (after `\usepackage{citemsp}`) so it is available throughout the document:

```
\usepackage{citemsp}
\citemspprefix{R}{Rem.}      "R" renders as "Rem."
\citemspprefix{w}{w.}       "w" renders as "w."
```

For example, after registering `R` for “Remark”: `\citemsp{cox1979/12.4/R3}` → $[4^{\S 12.4}_{\text{Rem. 3}}]$, and after registering `w` for “widget”: `\citemsp{zongker2006/3/w5}` → $[5^{\S 3}_{\text{w5}}]$. The prefix letter must not collide with an existing built-in prefix.

Prefix	Locator	Render	In super slot	In sub slot	Both slots
A	Appendix	App.	$[1^{\text{App.2}}]$	$[1_{\text{App.2}}]$	$[1^{\S 3.2}_{\text{App.2}}]$
C	Corollary	Cor.	$[1^{\text{Cor.3}}]$	$[1_{\text{Cor.3}}]$	$[1^{\S 4.1}_{\text{Cor.3}}]$
L	Lemma	Lem.	$[1^{\text{Lem.1}}]$	$[1_{\text{Lem.1}}]$	$[1^{\S 3.2}_{\text{Lem.1}}]$
T	Theorem	Th.	$[1^{\text{Th.4.2}}]$	$[1_{\text{Th.4.2}}]$	$[1^{\S 6.1}_{\text{Th.4.2}}]$
d	Definition	$\triangleq$	$[1^{\triangle 5}]$	$[1_{\triangle 5}]$	$[1^{\S 3.2}_{\triangle 5}]$
e	Equation	eq.	$[1^{\text{eq.4.3}}]$	$[1_{\text{eq.4.3}}]$	$[1^{\S 6.2}_{\text{eq.4.3}}]$
f	Figure	fig.	$[1^{\text{fig.3}}]$	$[1_{\text{fig.3}}]$	$[1^{\S 6.1}_{\text{fig.3}}]$
n	Footnote	†	$[3^{\dagger 7}]$	$[3_{\dagger 7}]$	$[3^{\S 1}_{\dagger 7}]$
p	Page	pp.	$[1^{\text{pp.42}}]$	$[1_{\text{pp.42}}]$	$[1^{\S 3.2}_{\text{pp.42}}]$
t	Table	$\boxplus$	$[2^{\boxplus 2}]$	$[2_{\boxplus 2}]$	$[2^{\S 8.4}_{\boxplus 2}]$

Table 1: Built-in prefix registry.

3.5 Configuration

Two user-configurable parameters control the locator appearance:

- `\citernpscale` (default 0.35) — controls the locator glyph size via `\scalebox` from `graphicx`.
- `\citernpraiseoffset` (default 1.5pt) — vertical offset applied via `\raisebox` to superscript locators.

Redefine them after loading:

```
\renewcommand{\citernpscale}{0.4}
\renewcommand{\citernpraiseoffset}{2pt}
```

4 Comparison with Standard Biblalex

Standard biblalex	citemsp
<code>\autocite[§3.2]{smith2003}</code> → [1, §3.2]	<code>\citemsp{smith2003/3.2}</code> → $[1^{\S 3.2}]$
<code>\autocites[§3.2]{smith2003}</code> <code>[eq.~4.3]{meyerrochow2003}</code> → [1, §3.2][2, eq. 4.3]	<code>\citemsp{smith2003/3.2,</code> <code>meyerrochow2003/e4.3}</code> → $[1^{\S 3.2}, 3^{\text{eq.4.3}}]$
<code>\autocites[§3.2, ¶5]{smith2003}</code> → [1, §3.2, ¶5]	<code>\citemsp{smith2003/3.2/5}</code> → $[1^{\S 3.2}_{\P 5}]$
No built-in typed locator syntax; user must type symbols manually	Ten built-in prefixes; extensible via <code>\citemspprefix</code>
Range compression via <code>numeric-comp</code>	Plain entries (no /) compressed via the backend's native <code>\cite</code> , so compression works identically

Table 2: Feature comparison between standard `biblalex` locators and `citemsp`.

4.1 Source-level comparison

The difference in source verbosity becomes clear when multiple locator-bearing citations appear together. Consider six sources:

```
% Standard biblatex (6 sources, ~185 characters):
\autocites[$3.2]{smith2003}[¶2]{meyerrochow2003}
[eq.\,6.55]{matthews1995}[$8.4, tab.\,2]{wang2020}
[$2.3]{ghirlanda2002}[$3]{zongker2006}

% citemsp equivalent (6 sources, ~114 characters):
\citemsp{smith2003/3.2, meyerrochow2003//2,
matthews1995/e6.55, wang2020/8.4/t2,
ghirlanda2002/2.3, zongker2006/3}
```

Rendered output:

biblatex: [1, §3.2, 3, ¶2, 6, eq. 6.55, 2, §8.4, tab. 2, 7, §2.3, 5, §3]

citemsp: [1^{§3.2}, 3_{¶2}, 6^{eq.6.55}, 2_{§8.4}, 7^{§2.3}, 5^{§3}]

At six entries the `biblatex` source is roughly 40% longer, and the rendered bracket sequence occupies significant horizontal space. The `citemsp` output remains a single compact bracket with superscript/subscript annotations.

5 Examples

5.1 Single source with section and paragraph

The lack of randomised controlled trials on
parachutes~\citemsp{smith2003/6.1/2} remains a gap in
evidence-based medicine.

Output: The lack of randomised controlled trials on parachutes [1_{¶2}^{§6.1}] remains a gap in evidence-based medicine.

5.2 Section-only locator

Chickens were found to prefer beautiful
humans~\citemsp{ghirlanda2002/2.3}, raising questions about
peer review in poultry science.

Output: Chickens were found to prefer beautiful humans [7^{§2.3}], raising questions about peer review in poultry science.

5.3 Subscript-only locator

An empty first slot (double slash) gives a subscript without a superscript:

```
\citemsp{smith2003//5}           % paragraph 5, no section
\citemsp{meyerrochow2003//e2.1} % equation 2.1, no section
```

Output: [1_{¶5}], [3_{eq.2.1}].

5.4 Multiple sources with mixed locators

```
Writer's block~\citemsp{upper1974, smith2003/4,  
  meyerrochow2003/1/2} has been studied from behavioural,  
medical, and ornithological perspectives.
```

Output: Writer's block [8, 1^{§4}, 3^{§1}₂] has been studied from behavioural, medical, and ornithological perspectives.

5.5 Range compression

Plain entries (no slashes) are batched and passed to the backend's native `\cite`, so range compression works automatically when using `numeric-comp` (biblatex) or `sort&compress` (natbib). Entries *with* locators are rendered individually and interleaved with the compressed group. The examples below illustrate different combinations.

1. All plain (fully compressed):

```
\citemsp{upper1974, pennycook2015, smith2003, meyerrochow2003,  
  wang2020}
```

Output: [8, 9, 1, 3, 2]

2. Plain entries mixed with locator entries:

```
\citemsp{upper1974, pennycook2015, smith2003/4.1,  
  meyerrochow2003, wang2020}
```

Output: [8, 9, 1^{§4.1}, 3, 2]

The plain entries compress into a range while the locator entry renders separately.

3. Multiple locator entries among plain ones:

```
\citemsp{upper1974, pennycook2015/2.3, smith2003,  
  meyerrochow2003/e14, wang2020}
```

Output: [8, 9^{§2.3}, 1, 3^{eq.14}, 2]

4. Every entry has a locator (no compression):

```
\citemsp{upper1974/1, smith2003/4.1/2,  
  meyerrochow2003/e14, wang2020/p21}
```

Output: [8^{§1}, 1^{§4.1}₂, 3^{eq.14}, 2^{pp.21}]

When every entry carries a locator, each is rendered individually and no range compression occurs.

5. Locator entries with super and subscript among plain ones:

```
\citemsp{upper1974, pennycook2015, smith2003/6.1/2,  
  meyerrochow2003, wang2020, cox1979/12.4/T5}
```

Output: [8, 9, 1^{§6.1}₂, 3, 2, 4^{§12.4}_{Th.5}]

5.6 Equation and figure locators

The sloshing dynamics of coffee~\citemsp{mayer2012/3.2/e3.31}
and the critical spill diagram~\citemsp{mayer2012/4.6/f4.1}
explain why your laptop is always sticky.

Output: The sloshing dynamics of coffee [10^{§3.2}_{eq.3.31}] and the critical spill diagram [10^{§4.6}_{fig.4.1}] explain why your laptop is always sticky.

5.7 Theorem, lemma, and corollary locators

The Cox--Zucker machine~\citemsp{cox1979/12.4/T5} computes
intersection numbers; see Lemma~\citemsp{cox1979/12.3/L2}
and Corollary~\citemsp{cox1979/12.4/C3}.

Output: The Cox--Zucker machine [4^{§12.4}_{Th.5}] computes intersection numbers; see Lemma [4^{§12.3}_{Lem.2}] and Corollary [4^{§12.4}_{Cor.3}].

5.8 Definition and page locators

The definition of a necrobotic
actuator~\citemsp{yap2022/2.3/d2.1} involves reanimating
dead spiders; see also~\citemsp{yap2022/p37}.

Output: The definition of a necrobotic actuator [11^{§2.3}_{2.1}] involves reanimating dead spiders; see also [11^{pp.37}].

5.9 Appendix and footnote locators

The theorem that biologists cannot fix a
radio~\citemsp{lazebnik2002/T11} is proven in
Appendix~\citemsp{lazebnik2002/A2}; the apology to
electrical engineers is in
footnote~\citemsp{lazebnik2002/3/n4}.

Output: The theorem that biologists cannot fix a radio [12^{Th.11}] is proven in Appendix [12^{App.2}]; the apology to electrical engineers is in footnote [12^{§3}₄].

5.10 Table locator and mixed types in one call

See~\citemsp{smith2003/6.1/2, matthews1995/e6.55,
wang2020/8.4/t2, yap2022/p170} for parachutes, toast,
graphene, and dead spiders respectively.

Output: See [1^{§6.1}₂, 6^{eq.6.55}, 2^{§8.4}₂, 11^{pp.170}] for parachutes, toast, graphene, and dead spiders respectively.

5.11 Plain citation as drop-in replacement

Without slashes, \citemsp behaves identically to \cite:

```
Elephant dung turns out to be useful
habitat~\citemsp{camposarceiz2009}, which is more than can be
said for most conference proceedings.
```

Output: Elephant dung turns out to be useful habitat [13], which is more than can be said for most conference proceedings.

5.12 Adjacent to inline math

```
The penguin defaecation pressure $P \approx 60\$,
\mathrm{kPa}$~\citemsp{meyerrochow2003/e14, smith2003/4.3}
exceeds most expectations.
```

Output: The penguin defaecation pressure $P \approx 60$ kPa [3^{eq.14}, 1^{§4.3}] exceeds most expectations.

5.13 Font contexts

The package works correctly in all standard L^AT_EX font contexts. For italic and slanted shapes, a small kern is automatically inserted to prevent the locator from colliding with the slanted closing bracket. Bold, bold-italic, small caps, and upright text work without modification.

```
Upright: [1§6.12]
Italic: [1§6.12]
Slanted: [1§6.12]
Bold: [1§6.12]
Bold italic: [1§6.12]
SMALL CAPS: [1§6.12]
Emph (italic): [1§6.12] — nested emph (upright): [1§6.12]
```

5.14 Font sizes

Locator glyphs scale proportionally at different font sizes:

```
Small: [7§2.3eq.2.5]
Normal: [7§2.3eq.2.5]
Large: [7§2.3eq.2.5]
```

5.15 Custom prefixes

Place `\citemsp{prefix}` in the preamble, after loading `citemsp`:

```
% in the preamble:
\citemsp{prefix}{w}{w.}      % "w" renders as "w."
\citemsp{prefix}{H}{Hyp.}    % "H" renders as "Hyp."

% in the document body:
\citemsp{zongker2006/3/w5}
\citemsp{cox1979/12.4/H2}
```

Output: $[5_{w.5}^{\S 3}]$, $[4_{Hyp.2}^{\S 12.4}]$

6 Natbib Backend

Since v2.2, `citemsp` auto-detects whether `biblatex` or `natbib` is loaded and selects the matching code path. No package option is needed.

6.1 Compatible classes

Many journal classes auto-load `natbib`. With these, simply loading `citemsp` after the class is sufficient:

Class	Backend	Notes
article, report, book, KOMA	either	user's choice
revtex4-2 (APS)	natbib (auto)	class loads natbib
aastex631 (AAS)	natbib (auto)	class loads natbib
elsarticle (Elsevier)	natbib (manual)	add <code>\usepackage[numbers]{natbib}</code>
IEEEtran natbib mode	natbib (manual)	add <code>\usepackage[numbers]{natbib}</code>

Table 3: Document class compatibility.

6.2 Minimal natbib example

```
\documentclass{article}
\usepackage[numbers, sort&compress]{natbib}
\usepackage{citemsp}
\begin{document}
See~\citemsp[key1/3.2/5, key2/e4.1] for details.
\bibliographystyle{unsrt}
\bibliography{refs}
\end{document}
```

On the `natbib` path, `citemsp` uses `\citenum` for the bare number and wraps it in a `\hyperlink` when `hyperref` is loaded. Range compression via `sort&compress` works for plain entries.

7 Implementation

- Backend detection.** At load time, the package checks whether `biblatex` or `natbib` is already loaded. If neither is present, a descriptive error is raised. If both are loaded (itself an error), then `biblatex` is preferred.
- Locator scaling.** `\citemspscale` (default 0.35) controls the size of locator glyphs via `\scalebox` from `graphicx`. `\citemspraiseoffset` (default 1.5pt) sets the vertical offset of superscript locators via `\raisebox`.
- Font shape detection.** The renderer checks `\f@shape` against `it` (italic) and `sl` (slanted). In either case, a small math kern is inserted to prevent the locator from colliding with the slanted final bracket.
- Prefix dispatch.** Each locator value is tested: if its first character matches a registered prefix (via `\csname @citemsp@pfx@...` lookup), the prefix symbol is rendered; otherwise the default $\S$ or ¶ is used.

5. **Citation rendering.** On the biblatex path, a custom cite command `\citelinksp` (via `\DeclareCiteCommand`) prints the hyperlinked number and attaches locators. On the natbib path, `\citenum` provides the bare number, wrapped in `\hyperlink` when `hyperref` is loaded.
6. **Entry parsing.** The user-facing `\citemsp` uses `\forcsvlist` (from `etoolbox`) to iterate over comma-separated entries. Each entry is split at forward slashes using a delimited `\def` pattern, extracting the key and up to two locator slots.
7. **Plain-key batching.** Entries without slashes are accumulated into a buffer and flushed via the backend’s native `\cite` (with brackets suppressed), so that range compression (`numeric-comp` or `sort&compress`) works identically to a regular `\cite` call.

8 Limitations

- **Numeric styles only.** `citemsp` attaches locators to numeric citation labels. Other non-numeric styles are currently not supported.
- **Manual paragraph counting.** Paragraph locators refer to paragraph numbers counted manually from the start of the referenced section. There is no automatic cross-referencing.
- **revtex4-2 with biblatex.** The combination of `revtex4-2` and `biblatex` is incompatible due to conflicts in `\addtocontents` patching between the two packages. Although not a `citemsp` issue, users should keep that in mind. Make sure to use the natbib backend with `revtex`.

9 Conventions

- **Section numbering:** use the source’s native format (e.g., 3.2.1, IV, A.1).
- **Paragraph counting:** count paragraphs from the start of the referenced section. This is a manual convention; `citemsp` does not verify paragraph numbers against the cited source.
Note: Paragraph locators rely on manual counting from the start of the referenced section. Since paragraphs are not explicitly numbered in published sources, these locators may be ambiguous across editions or between readers. When using ¶ locators, consider providing additional context in the surrounding text so the reader can locate the passage independently.
- **Subsection references:** use the full path (e.g., key/3.2.1), not a relative number.
- **Plain citations:** `\citemsp{key}` with no slashes produces a standard numeric citation, so the command can serve as a drop-in replacement for `\cite` if desired.

10 Requirements

- L^AT_EX2e (2020/10/01 or later)
- `biblatex` (any numeric style) **or** `natbib` (with the `numbers` option)
- `graphicx`
- `etoolbox`
- `amssymb` — recommended for best glyph quality ($\triangleq$, $\boxplus$). If not loaded, `citemsp` provides primitive fallback definitions.

11 License

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